

Math 170E

Week 5

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Example 1. (Final 20 Summer) Balin and Dwalin are two brothers, playing a game with a slightly crooked coin, with $P(H) = 0.6$ and $P(T) = 0.4$. The coin is tossed 10 times (each toss is independent from others) and in any turn it shows heads, it is tossed again. The brothers are counting the cases where the coin is tossed twice and the second toss, too, is heads. Here is an example scenario:

$$T(HT)TTTTTTT(HH)$$

The count here will be 1, as there are only two cases where the coin showed heads and in only one such case, the second toss was also heads. Thus, only the 10th turn is counted.

Let X be the counted number of such double heads. What is the support, pmf, expected value and variance of X ?

Example 2. (Quiz - 20 Summer) Let Z be the random variable with support $S_Z = \{1, 2, 3\}$ and pmf $f(z) = z/6$, for $z \in S_Z$.

1. Find the MGF (moment generating function) of Z , $M_Z(t)$.
2. Using $M_Z(t)$, calculate $E[Z^3]$.

Example 3. You throw a dice repeatedly. Let X be the number of throws until you get three 6s. What is the expected number throws?

Example 4. (Old Quiz Problem-PSI 2.6.10-modified) Red Rose Tea randomly began placing one of 25 English porcelain miniature figurine in a 100-bag box of tea, selecting from ten figurine in the American Heritage series. A customer, a big fan of George Washington, wants to have three copies of George Washington figurine; one for his collection, one to give to an acquaintance, and one for his own entertainment.

On the average, how many boxes of tea must be purchased by a customer to obtain three copies of George Washington figurine?

Example 5. Let $X \sim \text{Poiiss}(\lambda)$.

1. Compute EX
2. Compute $M(t) = E(E^{tX})$
3. Compute $\text{Var}(X)$

Example 6. On average, how many rolls we need to throw a fair dice to get all 6 outcomes?

- (A)6 (B)36 (C)12 (D)14.7 (E) $\frac{144}{7}$

Example 7. (Old MT problem) We have two fair, six-sided dice, one of which is red and the other of which is blue. We roll both dice.

1. What is the probability that at least one of the dice results in an odd number?
2. What is the probability that the result of the blue die is strictly less than the result of the red die?
3. Let A be the event that the sum of the rolls is even. Let B be the event that the results of both rolls are numbers ≥ 4 . Are A and B independent events? Justify your answer.

Example 8. (Old MT problem) Suppose that a coin is not fair so that the probability of obtaining a head is $p \in (0, 1)$.

1. On average, how many flips are needed to obtain a head?
2. Find the probability the first time obtaining a head is an even number. Your final answer must not be an infinite series.